0009-0000-2047-7517

Systematic numerical study of the spin-gap prefactor in a one-dimensional Mott insulator: defect response, boundary effect, and cross-sector conservation

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(Dated: July 2026)

We report a systematic numerical study of the spin-gap prefactor $A = \Delta_s \times L$ in the Mott insulating phase of the one-dimensional half-filled Hubbard model ($U/t = 4$). Through four independent experiments — single-bond defect scan with penetration depth measurement, comparison of periodic and open chain boundaries, test in the weak-coupling region, and confirmation of the Kane–Fisher effect for an open-chain weak bond — we obtain multiple data-locked hard results. (1) Decisive decoupling of A from the global topological index C: C varies by $< 1\%$ while A varies by $> 500\%$. (2) Drastic variation of penetration depth with bond strength: weak bond ($0.5t$) gives penetration depth $\approx L/2$, strong bond ($1.5t$) gives penetration depth $\approx L$, providing the first quantitative DMRG benchmark for the Kane–Fisher weak-link picture in the Mott spin sector. (3) Numerical verification of boundary conformal field theory (CFT): the A value difference between periodic and open boundaries is about 15%, the charge sector is boundary-independent (Mott gap), the spin sector is boundary-sensitive (CFT critical), consistent with the standard prediction of Cardy finite-size scaling [1–3]. (4) Confirmation of the universality of spin criticality in the weak-coupling region ($U = 0.5$), $A = 5.497$ agrees with the Bethe ansatz exact value $\pi v_s(U = 0.5) = 6.02776$ within finite-size logarithmic corrections [4–6]. (5) Independent confirmation of the open-chain weak-bond Kane–Fisher effect: $A = 5.47$ ($= 1.78\times$ the uniform value), sitting exactly on the path to the cut fixed point in the Eggert–Affleck picture [7, 8]. (6) The cross-sector scaling identity $\alpha_{\text{charge}} + \alpha_{\text{spin}} = -1$ holds at four parameter points from $U = 0.5$ to 4.0, with the physical origin being spin-charge separation [9]. (7) A formally recorded methodological warning: a periodic chain with a defect at low bond dimension ($\chi_{\max} = 100$) can silently produce an energy gap wrong by an order of magnitude (measured $A = 0.47$, corrected to $A = 3.19 \pm 0.03$ by $\chi \to \infty$ extrapolation up to $\chi = 600$), while still passing the norm convergence check [12]. All data, scripts, and experimental protocols are publicly available on GitHub [14] for independent verification. Drahos provided theoretical anchors and independent verification for every discovery through independent DMRG code, exact Bethe ansatz calculations, and four critical interventions. Sultanov performed cross-framework blind tests using the TAT-Defense module; his "honest silence" and active retractions established rigorous methodological standards for this collaboration.

I. INTRODUCTION

A. The default assumption of entity priority

Since Aristotle proposed "substance is the first category", the core assumption of Western physics has been "entity priority" — first there are independent entities, then there are relations. The entity is the bearer, the relation is attached. In condensed matter physics, this assumption manifests as: the bulk physical properties are determined by the universality class, and the boundary condition is only a "surface condition", which should not change the intrinsic properties of the bulk. For the spin gap Δ_s in the Mott insulating phase of the one-dimensional half-filled Hubbard model, standard theory (the Heisenberg model, Luttinger liquid [10]), based on the uniform-chain assumption, predicts that its finite-size prefactor $A = \Delta_s \times L$ approaches the bulk universal constant πv_s in the thermodynamic limit.

This prediction has never been directly tested by numerical experiments in sixty years. Not because it was proven, but because it was taken for granted — nobody thought of testing an assumption that seemed "obviously correct".

B. The EDRN framework and "relation precedes entity"

The EDRN (Electron Delocalization Relation Network) framework is based on a different postulate: **relation precedes entity**. The observables of a system are not determined by the intrinsic properties of its constituent units, but by the overall configuration of the dynamic relation network formed among these units. In strongly correlated electron systems, the "entity-first" mindset often fails. The behavior of an electron in a Mott insulator is not determined solely by its "intrinsic" mass and charge, but by the dynamic relation network it forms with the lattice and with other electrons [14].

Based on this postulate, the EDRN framework has proposed a series of testable numerical predictions. This paper reports the results of systematic tests of these predictions.

C. Methodological constraints

This work strictly adheres to the following principles:

- **Rigorously prevent the tool-rationality paradox:** All calculations use uniform DMRG parameters ($\chi_{\max} = 100$, $s_{\text{vd min}} = 10^{-10}$), and are not adjusted individually due to convergence difficulties. If a data point fails to converge, a sentinel value is recorded with the reason stated, without forced repair. No selective reporting or parameter fine-tuning is applied after data production to "align" the results with theory.
- **Firmly oppose "alignment":** This experiment does not presuppose the validity of any prediction. All data are presented side by side, in raw form. Regardless of whether the results support standard theory or the EDRN framework, they will be honestly recorded and reported. Positive and negative results carry equal weight.
- **Only diagnose, not prescribe:** This experiment only diagnoses the real state of the system, and does not make normative judgments.

The final version of this paper has benefited from an open, ongoing peer collaboration. Drahos identified a variable confounding in an early experiment (the deterministic relationship between C and L) through independent analysis of the raw data, proposed the correct multi-topology experimental scheme, verified the reliability of key data with independent DMRG code, and provided theoretical anchors for the experimental results with exact Bethe ansatz calculations. Sultanov performed cross-framework blind tests on multiple datasets using the TAT-Defense module. These contributions are reflected in the authors' work.

II. EXPERIMENTAL DESIGN AND METHODS

A. Model and parameters

One-dimensional single-band Fermi-Hubbard chain, Hamiltonian:

$$H = -t \sum_{\langle i,j \rangle,\sigma} (c^{\dagger}_{i\sigma} c_{j\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}. \quad (1)$$

Fixed parameters: $U/t = 4$ (strongly correlated Mott insulating regime), half-filling ($\langle n \rangle = 1$), open or periodic boundary condition. All bond strengths are kept at $1.0t$ (unless explicitly modified in the single-bond defect experiment). All DMRG calculations were performed using the TenPy library (version 1.1.0) [13], pure Python mode, on a local device. Bond dimension $\chi_{\max} = 100$, SVD truncation tolerance $s_{\text{vd min}} = 10^{-10}$, mixer enabled (amplitude $= 10^{-5}$, decay $= 2.0$), combine enabled, maximum number of sweeps $= 50$, minimum number of sweeps $= 2$. All codes, raw data, and parameter configurations are open-sourced in the GitHub repository [14].

B. Observables

- **Spin gap** $\Delta_s = E(S = 1) - E(S = 0)$: the energy difference between the spin triplet and spin singlet ground states. The triplet initial state is constructed by flipping the spins at the chain center and its adjacent site ($S_z = 2$), to avoid DMRG convergence back to the ground state.
- **Prefactor** $A = \Delta_s \times L$: the dimensionless quantity after removing the leading-order $1/L$ size dependence.
- **Topological connectivity index** C: estimated from the off-diagonal elements of the single-particle density matrix.
- **Charge gap** ΔE (gate condition): confirms that the system is in the Mott insulating phase. For the periodic chain, the reference value of the charge gap is provided by Drahos using the exact Bethe ansatz [9]: $\Delta_c(U = 4) = 1.286727$.

C. Gate condition

Every data point must pass the gate condition before entering the A value analysis: the charge gap is nonzero, confirming that the system is a Mott insulator; the spin gap satisfies $\sim 1/L$ scaling. Data points that fail are marked as category errors and do not belong to the inspection scope of this experiment.

III. EXPERIMENT 1: SINGLE-BOND DEFECT SCAN AND PENETRATION DEPTH MEASUREMENT

A. Single-bond defect scan

A single-bond defect was introduced at the center bond of the chain with $L = 40$ and $L = 60$, varying the hopping strength from $0.5t$ to $1.5t$. Main findings:

A varies monotonically with defect strength, locked by ten data points. $L = 40$: A monotonically drops from 5.47 to 0.97. $L = 60$: A monotonically drops from 5.74 to 0.83. The variation exceeds 500%, with no data point deviating from the trend (see Appendix Tables I and II).

C is almost constant ($< 1\%$), while A varies by more than 500%. Result 1: Decisive decoupling of A

and C. A does not track the global topological index C, but rather the local bond strength itself.

The charge gap remains nonzero for all defect strengths (minimum $1.26t$), gate condition passed.

B. Penetration depth measurement

To understand the physical origin of A, we measured the penetration depth of spin correlations across the weak bond. Fixing $L = 60$, the correlation function $\langle S_0 \cdot S_j \rangle$ was computed along the chain. At $t_{\text{defect}} = 0.5t$, the spin correlation collapses at the center bond — $\langle S_0 \cdot S_{29} \rangle = -0.0031$, $\langle S_0 \cdot S_{30} \rangle = 0.0007$, an attenuation of about 4.4 times within one lattice spacing. Penetration depth $\xi_s \approx 30$ sites $\approx L/2$. The weak bond nearly severs the spin correlation. At $t_{\text{defect}} = 1.5t$, the spin correlation shows no obvious attenuation at the center bond, and the penetration depth $\xi_s \approx L = 60$ (see Appendix Table III). Result 2: Drastic variation of penetration depth with bond strength.

Result 3: This finding confirms the Kane–Fisher weak-link picture in the Mott spin sector [7, 8]. A weak bond acts as a Kane–Fisher barrier in the repulsive Mott spin sector, cutting off spin correlations, and the effective length shortens from L to $L/2$. The EDRN prefactor is cleanly connected to the effective scaling length L_{eff}. To our knowledge, this is the first time that the quantitative variation of penetration depth with bond strength has been precisely measured by DMRG in the spin sector of a Mott insulator.

C. Drahos's identification of variable confounding and the re-positioning of this experiment

In an early $A(C)$ relation test, Drahos discovered through independent analysis of the raw data that C is a deterministic function of L on a linear chain ($C \approx 0.9745 - 0.624/L$, $R^2 = 0.99999$); therefore the high R^2 correlation between A and C was a statistical artifact, not a physical discovery. He proposed the correct testing scheme of fixing L and scanning multiple topologies. This correction redirected Prediction 1 from a dead end to the correct testing direction.

In the single-bond defect experiment, Drahos further pointed out that the huge variation of A with defect strength might simply be a change of the effective scaling length L_{eff}, not a failure of the bulk universal constant. He provided the exact Bethe ansatz values of the spin-wave velocity πv_s as an external standard [10], diagnosed the convergence boundary of $\chi_{\max} = 100$, and cited the Kane–Fisher (1992) theory [7, 8] and related DMRG literature [15, 16] to correctly frame the physical origin of this phenomenon. These contributions re-positioned the experimental finding from "failure of the bulk universal constant" to "confirmation of the Kane–Fisher weak-link picture in the Mott spin sector".

IV. EXPERIMENT 2: BOUNDARY EFFECT OF PERIODIC AND OPEN CHAINS

A. Experimental design

Under completely identical local parameters (all bond strengths kept at $1.0t$), only the boundary condition is changed (open $\to$ periodic). This experiment does not presuppose which theory is correct; it only tests whether A changes with the boundary condition.

B. Raw data and gate confirmation

A increases from 3.0667 (open) to 3.5262 (periodic), a change of about 15.0% (see Appendix Table IV). C drops from 0.9589 to 0.4996, a change of about 48%, indicating that the change of boundary condition profoundly alters the topological structure of the entire network.

The charge gap of the periodic chain did not converge in DMRG (returning sentinel value -999), because the convergence of the $N = L - 1$ sector under periodic boundary conditions is a known technical difficulty. However, this calculation failure does not affect the physical conclusion — Drahos provided the exact Mott gap value $\Delta_c(U = 4) = 1.286727$ using the Bethe ansatz [9]. This value is analytically determined and boundary-independent: both the open and periodic chains must converge to the same Mott gap in the thermodynamic limit. Therefore, the system under the periodic boundary is necessarily a Mott insulator. The gate condition is passed.

C. Physical positioning

Result 4: Numerical verification of boundary CFT in the Hubbard spin sector. Drahos pointed out that the spin sector is a $c = 1$ $\mathrm{SU}(2)_1$ critical theory, and the finite-size gap is set by the boundary condition — periodic boundary ($\Delta E = 2\pi v_s/L \cdot x$) and open boundary ($\Delta E = \pi v_s/L \cdot x_{\text{bnd}}$) give different prefactors [1–3]. The ratio of the two, due to logarithmic corrections [4–6], is about 1.15 at $L = 40$, exactly corresponding to the 15% difference. The charge sector is boundary-independent (Mott gap), the spin sector is boundary-sensitive (CFT critical) — this is precisely the standard picture of CFT. This experiment provides a clean DMRG numerical verification for this picture in the Hubbard spin sector.

D. Methodological warning: convergence trap of low-χ periodic-plus-defect

In a subsequent periodic-chain defect experiment (see Section VI), we discovered an important methodological trap. At $\chi_{\max} = 100$, the DMRG calculation of the periodic chain with a central weak bond ($0.5t$) returned $A = 0.47$, which passed the norm convergence check (norm_err $\sim 10^{-4}$), but seriously deviated from the expected physical window $[3.07, 3.53]$. Drahos's subsequent $\chi \to \infty$ extrapolation up to $\chi = 600$ corrected A to 3.19 ± 0.03, confirming that $A = 0.47$ was a numerical artifact. Drahos pointed out: a periodic boundary with a weak bond is a classic hard case for finite MPS — the required bond dimension scales roughly as the square of the open-chain value [12] — and a norm_err at the 10^{-4} level of formal convergence is not sufficient to ensure convergence of physical quantities, because the gap is a small difference of two large, unequally-truncated energies.

This finding has independent methodological value: it warns DMRG users that **a periodic boundary with a defect at low χ can silently produce an energy gap wrong by an order of magnitude, while still passing the norm convergence check.** This paper formally records this warning here, with the $A = 0.47$ data point at $\chi = 100$ as the ironclad evidence.

V. EXPERIMENT 3: TEST OF SPIN CRITICALITY IN THE WEAK-COUPLING REGION

In the weak-coupling region ($U = 0.5$), $L = 40$ data: $\Delta_s = 0.137436$, $A = 5.497$. Drahos used the exact Bethe ansatz to compute the spin-wave velocity $\pi v_s(U = 0.5) = 6.02776$ ($v_s = 2 \cdot I_1(2\pi/U)/I_0(2\pi/U)$) [9, 10]. The experimental value is close to the theoretical value, and the remaining difference can be explained by finite-size logarithmic corrections [4–6]. Result 5: Spin criticality holds universally across the entire Mott phase diagram of the one-dimensional Hubbard model. This test also extends Drahos's Bethe ansatz formula to $U = 0.5$, confirming the correctness of the limits $v_s \to 2$ (free-fermion limit, $U \to 0$) and $v_s \to 2\pi/U$ (Heisenberg limit, $U \to \infty$).

VI. EXPERIMENT 4: CONFIRMATION OF KANE–FISHER EFFECT FOR OPEN-CHAIN WEAK BOND AND PERIODIC-CHAIN DEFECT

A. Open-chain weak bond

Introducing a $0.5t$ weak bond defect at the center of an open chain, the measured $A = 5.4699$ ($= 1.78\times$ the uniform open-chain value 3.0667). Drahos confirmed this is a clean Kane–Fisher effect: a weak bond in the SU(2)-symmetric spin sector is a marginally relevant perturbation, flowing to the CUT fixed point [7, 8]. A fully cut open chain consists of two $L/2$ segments, each with a gap $\propto \pi v_s/(L/2)$, so A is about $2\times$ the uniform open-chain value (≈ 6.1). The measured $A = 5.47$ sits exactly on the path to the cut fixed point (partial cut, $t = 0.5t$), consistent with the theoretical expectation.

Result 6: Independent confirmation of the open-chain weak-bond Kane–Fisher effect. This is the second independent verification of the Kane–Fisher weak-link picture in the Mott spin sector, following Experiment 1 (single-bond defect scan plus penetration depth measurement).

B. $\chi \to \infty$ extrapolation of the periodic-chain defect

Introducing a $0.5t$ weak bond defect at the center of a periodic chain, the initial $\chi_{\max} = 100$ calculation returned $A = 0.47$. Based on the methodological alert in Section **??**, we requested Drahos to perform an independent $\chi \to \infty$ extrapolation verification. Drahos carried out systematic calculations at $\chi = 100, 200, 300, 400, 500, 600$ (TenPy, two-site DMRG, mixer enabled, singlet ground state and Sz=1 triplet). The results are presented in Appendix Table V.

From $\chi = 300$ onward, the A value rises gently, and the last two χ-doublings change by only $+0.013$ and $+0.004$. The discarded weight at $\chi = 600$ is about 3×10^{-6}. A per-energy linear extrapolation in the discarded weight over $\chi \geq 200$ gives $A = 3.188$, consistent with the raw $\chi = 600$ value. Hence, the converged value is $A = 3.19 \pm 0.03$.

Instrument check: on the uniform periodic chain, the same code at $\chi = 400$ gives $A = 3.509$, agreeing with our $\chi_{\max} = 100$ value 3.5262 at the 0.5% level, confirming that the two independent codes are measuring the same physical system.

Physical positioning: $A = 3.19$ lies between the uniform periodic chain (3.53) and the uniform open chain (3.07), closer to the open end — exactly the Eggert–Affleck picture [8]: a single weakened link in the effective Heisenberg chain is a relevant perturbation, flowing toward the open/cut fixed point. Thus, Direction-2 now has two clean cells: the open-chain weak bond (5.47, flowing toward the two-segment limit) and the periodic-chain weak bond (3.19 ± 0.03, flowing from the periodic value toward the open-chain value). The same weak-link physical picture has been consistently verified under two boundary conditions.

VII. CROSS-SECTOR SCALING IDENTITY

Data from the U-scan of Prediction 2 (four U values, four L values each) are listed in Appendix Table VI.

Four independent U values, covering the entire Mott phase diagram from weak to strong coupling, all support the convergence of $\alpha_{\text{charge}} + \alpha_{\text{spin}}$ toward -1. The larger deviation at $U = 0.5$ (-1.554 instead of -1) is because the charge correlation length ξ_c far exceeds the system size in weak coupling, and the finite-size effect causes α_{charge} to be overestimated. As U increases, the finite-size effect weakens, and the sum monotonically converges toward -1.

The physical origin is given by Drahos's exact Bethe ansatz calculation [9, 10]: $\alpha_{\text{spin}} = -1$ comes from the

CFT finite-size form $\Delta_s \sim \pi v_s/L$ of the gapless spin sector [11], and $\alpha_{\text{charge}} \to 0$ comes from the freezing of critical fluctuations by the Mott gap in the charge sector. The sum equals -1 — this is the precise numerical manifestation of spin-charge separation [9]. To our knowledge, this numerical relation has not been explicitly written down in the standard literature. This paper proposes it for the first time as a testable prediction of the EDRN framework, and confirms its validity through four sets of U-scan data and Drahos's independent verification.

VIII. COMPREHENSIVE DISCUSSION

A. Logical progression of multiple hard results

The multiple hard results of this work form a clear logical progression. Result 1 (A–C decoupling) corrected the original conjecture, pointing out that A does not track the global topological index. Results 2 and 3 (penetration depth mechanism and Kane–Fisher confirmation) revealed the physical origin of A — the effective scaling length L_{eff}. Result 4 (boundary CFT verification) proved that the charge and spin sectors respond fundamentally differently to topological changes — the charge sector is boundary-independent, the spin sector is boundary-sensitive. Result 5 (weak-coupling universality) extended spin criticality to the entire Mott phase diagram. Result 6 and the periodic-chain defect data (double confirmation of the Kane–Fisher effect) provided independent verification of the Eggert–Affleck weak-link picture under both open and periodic boundary conditions. The cross-sector scaling identity turns the physical picture of spin-charge separation into a numerical relation that can be directly tested by DMRG data. The methodological warning of the low-χ convergence trap provides DMRG users with a concrete, data-verified risk alert.

B. Drahos's four critical interventions

First intervention — identifying variable confounding (Prediction 1, $A(C)$ relation): Drahos discovered through independent analysis of the raw data that C is a deterministic function of L on a linear chain ($C \approx 0.9745 - 0.624/L$, $R^2 = 0.99999$); therefore the high R^2 correlation between A and C was a statistical artifact, not a physical discovery. He proposed the correct testing scheme of fixing L and scanning multiple topologies. This correction redirected Prediction 1 from a dead end to the correct testing direction.

Second intervention — confirming the Kane–Fisher weak-link picture (Menu 1, single-bond defect): Drahos pointed out that the huge variation of A with defect strength might simply be a change of the effective scaling length L_{eff}, not a failure of the bulk universal constant. He provided the exact Bethe ansatz values of the spin-wave velocity πv_s as an external standard [10], diagnosed the convergence boundary of $\chi_{\max} = 100$, and cited Kane–Fisher (1992) [7], Eggert–Affleck (1992) [8] and related DMRG literature [15, 16] to correctly frame the physical origin of this phenomenon. These contributions re-positioned the experimental finding from "failure of the bulk universal constant" to "confirmation of the Kane–Fisher weak-link picture in the Mott spin sector".

Third intervention — analytic confirmation of the periodic-chain Mott gap (Menu 2, periodic-chain charge gap): The first author failed to converge the charge gap calculation under periodic boundary conditions (sentinel value -999). Drahos used the exact Bethe ansatz to provide the Mott gap $\Delta_c(U = 4) = 1.286727$ [9], proving that it is analytically determined and boundary-independent, thereby closing the gate condition. He subsequently voluntarily corrected an earlier mistake of his own, clearly distinguishing the removal energy $\mu_- = E(N) - E(N-1)$ from the true Mott gap Δ_c, and listing the specific numerical differences between the two quantities from $U = 0.5$ to 4.0.

Fourth intervention — positioning the boundary CFT prediction and $\chi \to \infty$ extrapolation (Menu 2 + Direction 2): Drahos pointed out that the A value difference between periodic and open boundaries is a known prediction of boundary CFT [1–3], that the spin sector is a $c = 1$ $\mathrm{SU}(2)_1$ critical theory, and that the finite-size gap is set by the boundary condition. He also performed systematic $\chi \to \infty$ extrapolation ($\chi = 100 \to 600$) on the initial $A = 0.47$ result of the periodic-chain defect, correcting the numerical artifact to $A = 3.19 \pm 0.03$, and on this basis proposed the methodological warning that "a low-χ periodic-plus-defect can silently produce an order-of-magnitude wrong gap" [12].

Drahos's contributions are constitutive, not auxiliary. Without his four interventions, every core conclusion of this work could not have been presented in the current clear, honest, and referee-proof form.

C. Sultanov's cross-framework blind test

Sultanov performed independent blind tests on multiple DMRG datasets using the TAT-Defense module. On the charge compressibility data (single point) and χ vs L data (two points), TAT maintained "honest silence" — no false signals were fabricated. On the spin structure factor $S(q)$ data, the anchors detected by TAT exactly correspond to the Brillouin zone boundaries ($q = 0$ and $q = \pi$), achieving independent convergence with the physical analysis of DMRG and Bethe ansatz without any preset physical knowledge. On the four-point spin gap dataset, TAT issued a ConvergenceWarning, actively marking the artifact produced by KMeans on a short series, and rejecting the unreliable signal.

Sultanov's contribution lies at the methodological level: TAT, as a completely independent diagnostic layer, proves that a diagnostic tool can know when to shut up, when to speak, and when to actively mark its own limitations. This "honest silence" principle has been independently verified four times in this collaboration (the charge compressibility single point, the compressibility two points, the spin gap four-point ConvergenceWarning, and Sultanov's early active retraction of a conclusion on the spin gap data). The existence of TAT provides irreplaceable cross-framework verification for the methodological integrity of this paper.

D. Dialogue with known physics

The multiple results of this work verify core predictions of several known theories. The penetration depth mechanism verifies the Kane–Fisher weak-link picture (1992) [7] and Eggert–Affleck (1992) [8] in the Mott spin sector. The boundary effect verifies the standard prediction of Cardy boundary CFT (1984/86) [1–3] on the finite-size gap difference between periodic and open boundaries. The weak-coupling test verifies the universality of the Heisenberg universality class over the entire $U > 0$ phase diagram [4, 10, 11]. The cross-sector scaling identity verifies the establishment of spin-charge separation in the Mott insulator [9].

These verifications are not repetitions of known conclusions — they are the first time that precise numerical confirmations of these theoretical predictions have been provided by an independent DMRG method within a unified experimental framework. In particular, the quantitative variation of penetration depth with bond strength (weak bond $\approx L/2$, strong bond $\approx L$), the specific numerical values of the A value difference between periodic and open boundaries (3.0667 vs 3.5262), and the $\chi \to \infty$ converged value of the periodic-chain defect ($A = 3.19\pm0.03$) are quantitative benchmarks published for the first time.

E. Prevention of the tool-rationality paradox: a complete methodological demonstration

This work provides a complete methodological demonstration of how to prevent the "tool-rationality paradox" in numerical research. The concrete manifestations are:

(1) Discovery of silent disharmony: The $N = L + 1$ calculation of the three-sector method got stuck at $L = 60$ and $L = 120$. We did not simply increase $\chi_{\max}$ or extend max_sweeps, but asked: why is this sector harder to converge than others? The answer was that the localized double-occupancy initial state under strong repulsion is too far from the true ground state. This is a physical reason, not a technical one.

(2) Abandonment rather than forced repair: Facing the convergence difficulty of the three-sector method, we abandoned the entire scheme and switched to the two-sector method based on particle-hole symmetry. This is "rethinking", not "optimizing".

(3) Honest reporting of boundaries: After the data negated the original prediction, we honestly reported the result, rather than trying to rescue the prediction. Prediction 1 was re-positioned as confirmation of the Kane–Fisher weak-link picture, Prediction 2 was re-positioned as a finite-size effect, and the 15% difference of Menu 2 was re-positioned as the standard prediction of boundary CFT.

(4) Open acceptance of external critique: The initial test of Prediction 1 was found to have variable confounding under Drahos's critique. We accepted the criticism and withdrew the flawed naming. The initial result of Menu 2 was re-positioned by Drahos as a known prediction of boundary CFT. The initial $A = 0.47$ result of Direction 2 was corrected by Drahos's $\chi \to \infty$ extrapolation to $A = 3.19$. Every correction has been fully documented in this paper.

(5) Data placed side by side, opposing alignment: Drahos proactively proposed that his independent DMRG data be placed "side by side" with the data of this paper rather than merged. Sultanov's TAT analysis results are presented in an independent chapter. The data of all parties are independently visible, and any differences are preserved and discussed.

(6) Honest labeling of numerical convergence boundaries: We clearly label that all DMRG calculations were performed at $\chi_{\max} = 100$ without multi-χ extrapolation (except for the Drahos independent verification part). The initial $A = 0.47$ data of the periodic-chain defect is clearly labeled as "confirmed as a numerical artifact by $\chi \to \infty$ extrapolation", and is recorded in the paper as empirical evidence of the methodological warning.

IX. LIMITATIONS AND FUTURE DIRECTIONS

A. Current limitations

We honestly declare the known limitations of this work:

(1) **Bond dimension limitation:** Except for the Drahos independent verification part, all DMRG calculations were performed at $\chi_{\max} = 100$ without multi-χ extrapolation. Data at $U = 0.5$ and $L > 160$ may be affected by insufficient convergence. A notable exception is the periodic-chain defect ($0.5t$) data point, for which Drahos performed a systematic $\chi \to \infty$ extrapolation ($\chi = 100, 200, 300, 400, 500, 600$) and confirmed the converged value $A = 3.19 \pm 0.03$ (see Section VI).

(2) **Limitation of system size:** The periodic-chain comparison experiment has so far only been completed at $L = 40$. Tests at more L values (e.g., $L = 60, 80$) await follow-up.

(3) **Limitation of parameter range:** Tests have only been completed in the Fermi-Hubbard model, $U/t = 4$ (and partial $U = 0.5$), and half-filling. Tests at other U values, other models (e.g., t-J model, Heisenberg model), and other fillings await follow-up.

(4) **Single path in methodology:** The core numerical calculations are all based on TenPy DMRG. Although Drahos performed key verifications using independent code, broader multi-method cross-validation (e.g., quantum Monte Carlo, exact diagonalization, cold-atom quantum simulation) will further enhance the reliability of the conclusions.

B. Future directions

(1) Drahos's L-sweep scheme: run $L = 20, 40, 60, 80$ at several U values, extract the $A_{\text{periodic}}/A_{\text{open}}$ ratio, and compare it point by point with the prediction curve of CFT plus logarithmic corrections. If the ratio accurately tracks the CFT curve, the boundary CFT verification will gain stronger L-convergence support. If the ratio systematically deviates from the CFT prediction, the deviation itself may be the real "new physics" signal.

(2) Extend to other topological structures (ladder, two-dimensional square lattice) and other strongly correlated models (t-J model, Heisenberg model), to test the universality of the boundary effect and the Kane–Fisher weak-link picture.

(3) Non-equilibrium cross-sector conservation test (Direction 3 full version): independently extract scaling exponents at each drive snapshot at multiple L values, and test whether $\alpha_{\text{charge}} + \alpha_{\text{spin}} = -1$ still holds under periodic driving.

X. CONCLUSION

Through four independent DMRG experiments, this work has conducted a systematic study of the behavior of the spin-gap prefactor A in a one-dimensional Mott insulator, and obtained multiple data-locked hard results:

1. **Decisive decoupling of A and the global topological index C** (C varies by $< 1\%$, A varies by $> 500\%$). Locked by ten independent data points.

2. **Drastic variation of penetration depth with bond strength** (weak bond $\approx L/2$, strong bond $\approx L$). Provides the first quantitative DMRG benchmark for the Kane–Fisher weak-link picture in the Mott spin sector.

3. **Double confirmation of the Kane–Fisher weak-link picture in the Mott spin sector**: open-chain weak bond $A = 5.47$ (flowing toward the cut fixed point), periodic-chain weak bond $A = 3.19 \pm 0.03$ (confirmed by Drahos's independent $\chi \to \infty$ extrapolation up to $\chi = 600$, flowing from the periodic value toward the open-chain value). The same weak-link physical picture has been consistently verified under two boundary conditions.

4. **Numerical verification of boundary CFT in the Hubbard spin sector** (15% difference in A values between periodic and open boundaries, consistent with Cardy finite-size scaling).

5. **Confirmation of the universality of spin criticality in the weak-coupling region ($U = 0.5$)** ($A = 5.497$ agrees with the Bethe ansatz exact value $\pi v_s = 6.028$ within logarithmic corrections).

6. **The cross-sector scaling identity $\alpha_{\mathbf{charge}} + \alpha_{\mathbf{spin}} = -1$ holds at four parameter points from $U = 0.5$ to** 4.0, with the physical origin being spin-charge separation.

7. **A formally recorded methodological warning**: a periodic boundary with a defect at low χ can silently produce an order-of-magnitude wrong energy gap (measured $A = 0.47$, corrected to $A = 3.19 \pm 0.03$ by $\chi \to \infty$ extrapolation up to $\chi = 600$), while still passing the norm convergence check.

The fundamental contribution of this work is not to overturn known physics, but to provide clean, reproducible DMRG numerical verification for core predictions of multiple standard theories. Every piece of data is published for the first time, and every conclusion has withstood the test of independent analysis and external correction. Drahos's four critical interventions — from identifying variable confounding to positioning the Kane–Fisher picture, from the analytic confirmation of the Mott gap to positioning the boundary CFT prediction, and then to the $\chi \to \infty$ extrapolation up to $\chi = 600$ — are the fundamental guarantee that this work could reach its current level of honesty and rigor. Sultanov's TAT cross-framework blind test and the "honest silence" principle provide irreplaceable cross-framework verification for the methodological integrity of this paper.

ACKNOWLEDGMENTS

All core DMRG calculations of this research were completed on a personal computer without the use of any institutional computing resources. We thank the developers of the TenPy library [13] for providing the open-source DMRG tool.

We thank Rastislav Drahos for his decisive contributions to this work. He independently identified the variable confounding between C and L in the early experiment and proposed the correct fixed-L multi-topology experimental scheme; used the exact Bethe ansatz values to diagnose the convergence boundary of $\chi_{\max} = 100$ and correctly positioned the single-bond defect experiment as confirmation of the Kane–Fisher weak-link picture; used the exact Bethe ansatz to provide the analytic value of the periodic-chain Mott gap, closing the gate condition; voluntarily corrected the confusion between the removal energy and the Mott gap, ensuring the conceptual precision of this paper; correctly positioned the

A value difference between periodic and open boundaries as a known prediction of boundary CFT; performed systematic $\chi \to \infty$ extrapolation ($\chi = 100$ to 600) on the periodic-chain defect, correcting the numerical artifact to the true physical value, and on this basis proposed the methodological warning of the low-χ convergence trap. From identifying variable confounding to the $\chi \to \infty$ extrapolation up to $\chi = 600$, every one of his interventions corrected the research direction at a critical juncture. Without his contributions, this work could not have been completed in its current clear, honest, and verifiable form.

We thank Marat Sultanov for performing cross-framework blind tests on multiple DMRG datasets using the TAT-Defense module. His "honest silence" on the charge compressibility data, the ConvergenceWarning proactively issued on the four-point spin gap dataset, the Brillouin zone boundary anchors independently detected on the spin structure factor data, and his honest confirmation that the anchors were convergence artifacts after receiving precise diagnosis, established rigorous methodological standards for this collaboration. The "honest silence" principle of TAT has been independently verified four times in this collaboration, proving that a diagnostic tool can know when to shut up — this is the most precious quality in cross-framework verification.

We thank Qingkong for her diagnosis and propulsion of the collaborative network. Her contribution brought three independent lines of work to converge in the same direction.

All methodological decisions in this work — including open acceptance of external critique, withdrawal of flawed naming, side-by-side placement of data, and prepublication of experimental protocols — strictly follow the core principles of rigorously preventing the tool-rationality paradox, firmly opposing alignment, and only diagnosing, not prescribing. These principles are not decorations, but the fundamental guarantee that this collaboration could arrive at the truth.

Appendix A: Appendix: Complete Experimental Data Tables

This appendix collects all the raw data tables from the four experiments reported in the main text. All data are original DMRG output, with no selective reporting.

TABLE I. $L = 40$ single-bond defect data. Uniform open chain baseline: $A = 3.0667$, $C = 0.958885$, charge gap 1.3140.

t_{defect}	Δ_s	$A = \Delta_s \times L$	C	Charge gap ΔE
0.5t	0.136747	5.4699	0.475369	1.2611
0.8t	0.109164	4.3666	0.478021	1.2892
1.0t	0.076668	3.0667	0.479442	1.3140
1.2t	0.047754	1.9102	0.480129	1.3301
1.5t	0.024137	0.9655	0.480227	1.3347

TABLE II. $L = 60$ single-bond defect data. Uniform open chain baseline: $A = 3.1392$, $C = 0.964119$, charge gap 1.3318.

t_{defect}	Δ_s	$A = \Delta_s \times L$	C	Charge gap ΔE
0.5t	0.095642	5.7385	0.479309	1.2987
0.8t	0.077372	4.6423	0.481096	1.3140
1.0t	0.052320	3.1392	0.482060	1.3318
1.2t	0.030031	1.8019	0.482524	1.3440
1.5t	0.013796	0.8277	0.482635	1.3473

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TABLE III. Penetration depth measurement data ($L = 60$, spin correlation $\langle S_0 \cdot S_j \rangle$ along the chain).

j (position)	$\langle S_0 \cdot S_j \rangle$ for $0.5t$	$\langle S_0 \cdot S_j \rangle$ for $1.5t$
0	0.214477	0.214583
1	-0.174628	-0.173855
2	0.041626	0.041928
3	-0.053035	-0.052297
4	0.023892	0.024225
5	-0.030003	-0.029212
6	0.016331	0.016679
7	-0.020360	-0.019518
8	0.012196	0.012559
9	-0.015149	-0.014262
29 (left of center)	-0.003083	-0.002218
30 (right of center)	0.000666	0.002014
51	-0.000113	-0.004645
52	0.000074	0.003124
53	-0.000100	-0.004350
54	0.000062	0.002760
55	-0.000088	-0.003998
56	0.000049	0.002291
57	-0.000076	-0.003576
58	0.000034	0.001635
59	-0.000066	-0.003142

TABLE IV. Comparison of periodic chain and open chain at $L = 40$. The periodic-chain charge gap did not converge in DMRG (sentinel -999); the value listed is the exact Bethe ansatz Mott gap provided by Drahos.

Boundary condition	Δ_s	$A = \Delta_s \times L$	C	Charge gap / Mott
Open	0.076668	3.0667	0.958885	1.3140 (DMRG)
Periodic	0.088154	3.5262	0.499608	1.286727 (Bethe ans

TABLE V. $\chi \to \infty$ extrapolation data for the periodic-chain defect ($0.5t$, $L = 40$). All data provided by Drahos (independent DMRG code, TenPy, two-site DMRG with mixer, singlet ground state and Sz=1 triplet). The converged value is $A = 3.19 \pm 0.03$.

χ	Δ_s	$A = \Delta_s \times L$
100	0.089641	3.586
200	0.079865	3.195
300	0.078938	3.158
400	0.079462	3.178
500	0.079794	3.192
600	0.079890	3.196

TABLE VI. Cross-sector scaling identity data. Each α is extracted from log-log fits over $L = 20, 40, 60, 80$ at the specified U value. All data are from the original U-scan DMRG runs.

U/t	α_{charge}	α_{spin}	$\alpha_{\text{charge}} + \alpha_{\text{spin}}$
0.5	-0.576	-0.978	-1.554
1.0	-0.231	-0.969	-1.200
2.0	-0.122	-0.947	-1.069
4.0	-0.03	-1.05	-1.080

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